

torch.nn.functional.max_pool3d#

https://pytorch.org/docs/stable/generated/torch.nn.functional.max_pool3d.html

Description:

Applies a 3D max pooling over an input signal composed of several input planes. The order of `ceil_mode` and `return_indices` is different from what seen in `MaxPool3d`, and will change in a future release. See `MaxPool3d` for details. `input` – input tensor (minibatch,in_channels,iD,iH,iW) (`\text{minibatch}` , `\text{in\_channels}` , `iD` , `iH` , `iW`) (minibatch,in_channels,iD,iH,iW), minibatch dim optional.

Function Signature

```
torch.nn.functional.max_pool3d(input, kernel_size,
stride=None, padding=0, dilation=1, ceil_mode=False,
return_indices=False)[source]#
```

Parameters

Notes & Warnings

NOTE

Note The order of `ceil_mode` and `return_indices` is different from what seen in `MaxPool3d`, and will change in a future release.

Detailed Documentation

Documentation

Applies a 3D max pooling over an input signal composed of several input planes.

The order of `ceil_mode` and `return_indices` is different from what seen in `MaxPool3d`, and will change in a future release.

See `MaxPool3d` for details.

`input` – input tensor (`minibatch, in_channels, iD, iH, iW`) (\text{minibatch} , \text{in_channels} , iD, iH , iW)(minibatch, in_channels, iD, iH, iW), minibatch dim optional.

`kernel_size` – size of the pooling region. Can be a single number or a tuple (`kT, kH, kW`)

`stride` – stride of the pooling operation. Can be a single number or a tuple (`sT, sH, sW`).

Default: `kernel_size`

`padding` – Implicit negative infinity padding to be added on both sides, must be ≥ 0 and $\leq \text{kernel_size} / 2$.

`dilation` – The stride between elements within a sliding window, must be > 0 .

`ceil_mode` – If `True`, will use ceil instead of floor to compute the output shape. This ensures that every element in the input tensor is covered by a sliding window.

`return_indices` – If `True`, will return the argmax along with the max values. Useful for `torch.nn.functional.max_unpool3d` later
